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JEE Main Physics Practice Sheet — DPP-044

Topic: Elastic Collisions in One Dimension — Velocity Formulations, Kinetic Energy Transfer & Successive Collisions**Time Allowed:** 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

- Q1.** A particle of mass m_1 moving with velocity u_1 undergoes a head-on perfectly elastic collision with a stationary particle of mass m_2 ($u_2 = 0$). The velocity v_1 of the first particle after the collision is:
- (A) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1$
 (B) $\left(\frac{2m_1}{m_1 + m_2}\right) u_1$
 (C) $\left(\frac{2m_2}{m_1 + m_2}\right) u_1$
 (D) $\left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_1$
- Q2.** In Question 1, the velocity v_2 of the second target particle after the collision is:
- (A) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1$
 (B) $\left(\frac{2m_1}{m_1 + m_2}\right) u_1$
 (C) $\left(\frac{2m_2}{m_1 + m_2}\right) u_1$
 (D) $\left(\frac{m_1 + m_2}{2m_1}\right) u_1$
- Q3.** Two identical billiard balls A and B of equal mass m collide head-on elastically. If A was moving with velocity $+6$ m/s and B was moving with velocity -4 m/s before the collision, their respective velocities after the collision are:
- (A) $+4$ m/s and -6 m/s
 (B) -4 m/s and $+6$ m/s
 (C) $+1$ m/s and $+1$ m/s
 (D) 0 m/s and $+2$ m/s
- Q4.** A moving ball of mass m strikes a stationary target of mass M head-on elastically. What is the mass ratio M/m such that the maximum possible fraction (100%) of initial kinetic energy is transferred to the target?
- (A) $M/m = 2$
 (B) $M/m = 0.5$
 (C) $M/m = 1$
 (D) $M/m \rightarrow \infty$
- Q5.** A neutron of mass m moving with kinetic energy K collides head-on elastically with a stationary carbon nucleus of mass $12m$. The fraction of the initial kinetic energy retained by the neutron is:
- (A) $\frac{121}{169}$
 (B) $\frac{169}{48}$
 (C) $\frac{169}{11}$
 (D) $\frac{13}{144}$
- Q6.** A heavy truck of mass $M = 2000$ kg moving at $v = 20$ m/s hits a small stationary rubber ball of mass $m = 0.1$ kg head-on elastically. The speed of the ball immediately after the collision is:
- (A) 20 m/s
 (B) 40 m/s
 (C) 10 m/s
 (D) 60 m/s
- Q7.** A light ball of mass m moving with speed u hits a massive rigid wall of mass $M \gg m$ moving towards it with speed v head-on elastically. The rebound speed of the light ball relative to the ground is:
- (A) $u + v$
 (B) $u + 2v$
 (C) $2u + v$
 (D) $u - v$
- Q8.** In a 1D elastic collision between two particles of masses m_1 and m_2 , the coefficient of restitution e is:
- (A) $e = 0$
 (B) $0 < e < 1$
 (C) $e = 1$
 (D) $e = \infty$
- Q9.** Three identical spheres A , B , and C of mass m are placed in a line on a smooth horizontal surface. Spheres B and C are initially at rest and in contact with each other. Sphere A strikes B head-on with velocity v . After all possible elastic collisions cease:
- (A) A and B move together with speed $v/2$
 (B) A and B come to rest, and C moves with velocity v
 (C) All three spheres move together with velocity $v/3$
 (D) A rebounds with $-v$, and B, C move forward with $v/2$
- Q10.** A body of mass m_1 moving at speed u_1 collides elastically and head-on with a stationary body of mass m_2 . The fractional loss in kinetic energy of the incident body ($\Delta K/K_i$) is:
- (A) $\frac{4m_1m_2}{(m_1 + m_2)^2}$
 (B) $\frac{(m_1 - m_2)^2}{(m_1 + m_2)^2}$
 (C) $\frac{2m_1m_2}{(m_1 + m_2)^2}$
 (D) $\frac{4(m_1 - m_2)^2}{(m_1 + m_2)^2}$
- Q11.** A particle of mass m moving with velocity u strikes a stationary particle of mass $2m$ head-on elastically. After the collision, the ratio of the speed of mass m to that of mass $2m$ is:
- (A) $1 : 2$
 (B) $2 : 1$
 (C) $1 : 1$
 (D) $1 : 4$
- Q12.** A ball of mass m moving with speed v_0 collides elastically with a stationary spring-block system (mass M attached to a spring of constant k). If $m = M$, the maximum compression of the spring is:

- (A) $v_0 \sqrt{\frac{M}{k}}$
 (B) $v_0 \sqrt{\frac{M}{2k}}$
 (C) $v_0 \sqrt{\frac{2M}{k}}$
 (D) $\frac{v_0}{2} \sqrt{\frac{M}{k}}$

Q13. In a one-dimensional elastic collision of two bodies, which of the following quantities is conserved during the entire deformation and restitution process at every instant?

- (A) Total kinetic energy only
 (B) Total linear momentum and total mechanical energy
 (C) Total kinetic energy and velocity of separation
 (D) Individual momentum of each body

Q14. A body A of mass M moving with velocity V collides elastically with a stationary body B . If body A rebounds with a velocity of $-V/3$, the mass of body B is:

- (A) $M/2$
 (B) $2M$
 (C) $3M$
 (D) $4M$

Q15. A ball P of mass m moving at 10 m/s collides elastically with another stationary ball Q . After collision, P continues moving in the forward direction with speed 2 m/s. The mass of ball Q is:

- (A) $\frac{2}{3}m$
 (B) $\frac{3}{2}m$
 (C) $4m$
 (D) $0.5m$

Q16. A particle of mass m_1 moving with speed $u_1 = 5$ m/s collides head-on elastically with another particle of mass $m_2 = 3$ kg moving in the same direction with speed $u_2 = 2$ m/s. If $m_1 = 1$ kg, the velocity of m_1 after the collision is:

- (A) +3.5 m/s

- (B) +0.5 m/s
 (C) -0.5 m/s
 (D) +2.0 m/s

Q17. In Question 16, the velocity of the 3 kg particle after the collision is:

- (A) +3.5 m/s
 (B) +4.0 m/s
 (C) +2.5 m/s
 (D) +1.5 m/s

Q18. Two masses m and M undergo a 1D elastic collision. At the instant of maximum deformation (maximum potential energy of deformation):

- (A) The relative velocity between them is zero (both move at v_{cm})
 (B) The total kinetic energy of the system is zero
 (C) The kinetic energy equals the initial kinetic energy
 (D) The acceleration of both bodies is zero

Q19. In Question 18, if mass m was initially moving with velocity u and M was at rest, the maximum potential energy stored at the moment of maximum deformation is:

- (A) $\frac{1}{2} \left(\frac{Mm}{M+m} \right) u^2$
 (B) $\frac{1}{2} mu^2$
 (C) $\frac{1}{2} \left(\frac{m^2}{M+m} \right) u^2$
 (D) $\frac{1}{4} mu^2$

Q20. A ball of mass m falls from height h onto a smooth horizontal floor. If the collision is perfectly elastic ($e = 1$), the time period of the periodic bouncing motion is:

- (A) $\sqrt{\frac{2h}{g}}$
 (B) $2\sqrt{\frac{2h}{g}}$
 (C) $4\sqrt{\frac{2h}{g}}$
 (D) $\pi\sqrt{\frac{h}{g}}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-044)

Elastic Collisions in One Dimension — Velocity Formulations & Energy Transfers

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	B	11	A	16	B
2	B	7	B	12	A	17	A
3	B	8	C	13	B	18	A
4	C	9	B	14	B	19	A
5	A	10	A	15	A	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)

From conservation of momentum and $e = 1$:

$$m_1 u_1 = m_1 v_1 + m_2 v_2 \quad \text{and} \quad v_2 - v_1 = u_1$$

Substituting $v_2 = u_1 + v_1$ into momentum equation:

$$m_1 u_1 = m_1 v_1 + m_2 (u_1 + v_1) \implies (m_1 + m_2) v_1 = (m_1 - m_2) u_1$$

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1$$

Sol 2. Option (B)

Using $v_2 = u_1 + v_1$:

$$v_2 = u_1 + \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1$$

Sol 3. Option (B)

When two bodies of equal mass ($m_1 = m_2$) collide elastically in 1D, they exchange their velocities completely:

$$v_A = u_B = -4 \text{ m/s}, \quad v_B = u_A = +6 \text{ m/s}$$

Sol 4. Option (C)

Fractional energy transferred to the target:

$$\frac{K_2}{K_1} = \frac{\frac{1}{2} M v_2^2}{\frac{1}{2} m u^2} = \frac{M}{m} \left(\frac{2m}{m+M} \right)^2 = \frac{4mM}{(m+M)^2}$$

This expression achieves its maximum value of 1 (100%) when $m = M \implies M/m = 1$.

Sol 5. Option (A)

Velocity of neutron after collision:

$$v = \left(\frac{m - 12m}{m + 12m} \right) u = -\frac{11}{13} u$$

Fraction of kinetic energy retained:

$$\frac{K'}{K} = \frac{\frac{1}{2} m v^2}{\frac{1}{2} m u^2} = \left(-\frac{11}{13} \right)^2 = \frac{121}{169}$$

Sol 6. Option (B)

Since $M \gg m$, $v_2 \approx \left(\frac{2M}{M+m} \right) v \approx 2v$.

$$v_2 = 2(20) = 40 \text{ m/s}$$

Sol 7. Option (B)

Relative velocity of approach is $u_{\text{rel}} = u - (-v) = u + v$.

For elastic collision ($e = 1$), relative velocity of separation equals approach:

$$v_{\text{rel}} = v_{\text{ball}} - v_{\text{wall}} = u + v \implies v_{\text{ball}} = v + (u + v) = u + 2v$$

Sol 8. Option (C)

By definition, a perfectly elastic collision preserves kinetic energy before and after collision without any permanent deformation, giving $e = 1$.

Sol 9. Option (B)

A strikes stationary B : A comes to rest and transfers speed v to B .

B then immediately strikes stationary C : B comes to rest and transfers speed v to C .

Final state: A and B are stationary at rest, and C moves forward with speed v .

Sol 10. Option (A)

Fractional loss of kinetic energy from m_1 :

$$\frac{\Delta K}{K_i} = 1 - \frac{K_f}{K_i} = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 = \frac{(m_1 + m_2)^2 - (m_1 - m_2)^2}{(m_1 + m_2)^2} = \frac{4m_1 m_2}{(m_1 + m_2)^2}$$

Sol 11. Option (A)

Velocity of m : $v_1 = \left(\frac{m - 2m}{m + 2m} \right) u = -\frac{1}{3} u \implies |v_1| = \frac{1}{3} u$.

Velocity of $2m$: $v_2 = \left(\frac{2m}{m + 2m} \right) u = \frac{2}{3} u$.

Ratio of speeds:

$$\frac{|v_1|}{v_2} = \frac{u/3}{2u/3} = \frac{1}{2}$$

Sol 12. Option (A)

Since $m = M$, the incident ball stops completely, transferring its entire velocity v_0 to block M .

Mechanical energy conservation for spring-block:

$$\frac{1}{2} M v_0^2 = \frac{1}{2} k x_{\text{max}}^2 \implies x_{\text{max}} = v_0 \sqrt{\frac{M}{k}}$$

Sol 13. Option (B)

Because no external forces act on the colliding system, total linear momentum is conserved at every moment. Total mechanical energy ($K + U_{\text{elastic}}$) is also conserved throughout.

Sol 14. Option (B)

Velocity of A after collision:

$$v_A = \left(\frac{M - M_B}{M + M_B} \right) V = -\frac{V}{3}$$

$$\frac{M - M_B}{M + M_B} = -\frac{1}{3} \implies 3M - 3M_B = -M - M_B \implies 4M = 2M_B \implies M_B = 2M$$

Sol 15. Option (A)

Velocity formula:

$$v_P = \left(\frac{m - m_Q}{m + m_Q} \right) (10) = 2 \implies \frac{m - m_Q}{m + m_Q} = \frac{1}{5}$$

$$5m - 5m_Q = m + m_Q \Rightarrow 6m_Q = 4m \Rightarrow m_Q = \frac{2}{3}m$$

Sol 16. Option (B)

Using general formula:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2$$

$$v_1 = \left(\frac{1-3}{1+3} \right) (5) + \left(\frac{2 \times 3}{1+3} \right) (2) = \left(-\frac{2}{4} \right) (5) + \left(\frac{6}{4} \right) (2) = -2.5 + 3.0 = +0.5 \text{ m/s}$$

Sol 17. Option (A)

Velocity of m_2 :

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2$$

$$v_2 = \left(\frac{2 \times 1}{4} \right) (5) + \left(\frac{3-1}{4} \right) (2) = \frac{10}{4} + \frac{4}{4} = \frac{14}{4} = +3.5 \text{ m/s}$$

Sol 18. Option (A)

Maximum deformation occurs when the two bodies are at their closest approach, which occurs when their relative velocity momentarily drops to zero ($v_{\text{rel}} = 0$), so both travel at the center-of-mass velocity v_{cm} .

Sol 19. Option (A)

Velocity of CM: $v_{\text{cm}} = \frac{mu}{m+M}$.

Kinetic energy at maximum deformation:

$$K_{\text{cm}} = \frac{1}{2}(m+M)v_{\text{cm}}^2 = \frac{1}{2} \frac{m^2 u^2}{m+M}$$

Stored deformation potential energy:

$$U_{\text{max}} = K_i - K_{\text{cm}} = \frac{1}{2}mu^2 - \frac{1}{2} \frac{m^2 u^2}{m+M} = \frac{1}{2} \left(\frac{Mm}{m+M} \right) u^2$$

Sol 20. Option (B)

Speed before impact: $v = \sqrt{2gh}$. Rebound speed: $v' = v = \sqrt{2gh}$.

Time to fall: $t_1 = \sqrt{\frac{2h}{g}}$. Time to rise: $t_2 = \sqrt{\frac{2h}{g}}$.

Total period of one bounce:

$$T = t_1 + t_2 = 2\sqrt{\frac{2h}{g}}$$